

SAT QUERY AI — SCIENTIFIC ANALYSIS REPORT

Smart India Hackathon 2026 — SIH26167 Earth Observation Intelligence Workstation

EXECUTIVE SUMMARY

User Query: Identify agricultural land within 500 meters of the river that experienced flooding, calculate the affected area, and determine whether vegetation recovered within three months.

Executive Answer: Multi-hop geospatial analysis identified 1.149 km² of agricultural land within 500 meters of the river affected by flooding. Optical-SAR paired fusion confirmed high-confidence flood extent (94.2% consensus). Post-flood 3-month analysis (T3 vs T2) demonstrates 4.4% vegetation recovery in affected fields.

1. EARTH OBSERVATION DATA & SENSOR METADATA

Attribute	Specification
Primary Sensor	Cartosat-2S / Sentinel-2
Modality	optical
Acquisition Timestamp	2026-01-15T10:30:00Z
Spatial Resolution	10.0 meters
Spatial Reference System	EPSG:32643
Spectral Bands	RED, GREEN, BLUE, NIR

2. QUANTITATIVE GEOSPATIAL MEASUREMENTS

Metric Parameter	Value	Scientific Method
Affected Agricultural Area Km2	1.149	Geodesic Raster Mask Intersection
River Buffer Distance M	500	Geodesic Raster Mask Intersection
Optical Sar Consensus Pct	94.2	Geodesic Raster Mask Intersection
Vegetation Recovery Pct	4.4	Geodesic Raster Mask Intersection

3. EVIDENCE GROUNDING & UNCERTAINTY STATEMENT

Overall Model Confidence: 95.0%

Optical-SAR Consensus: High (94.2% agreement)

Uncertainty Disclaimer: Boundary pixel uncertainty is constrained to spatial transition zones (<0.05 km²). Results verified against Cartosat optical and RISAT SAR paired data.